

```
import math canal_I = [] canal_I.append(0.82) canal_I.append(0.79) canal_I.append(0.85)
canal_I.append(5.20) canal_I.append(0.81) canal_I.append(0.78) canal_I.append(0.83)
canal_Q = [] canal_Q.append(0.41) canal_Q.append(0.39) canal_Q.append(0.43)
canal_Q.append(0.40) canal_Q.append(-4.80) canal_Q.append(0.42) canal_Q.append(0.38)
sinal = canal_I.copy() sinal.extend(canal_Q) sinal_backup = sinal.copy() media = sum(sinal) /
len(sinal) soma_quadrados = sum((x - media) ** 2 for x in sinal) desvio_padrao =
math.sqrt(soma_quadrados / len(sinal)) limite_inferior = media - 3 * desvio_padrao
limite_superior = media + 3 * desvio_padrao ruidos = [] for x in sinal: if x < limite_inferior or x >
limite_superior: ruidos.append(x) for r in ruidos: sinal.remove(r) print('Sinal bruto (',
len(sinal_backup), 'amostras ):', sinal_backup) print('Ruídos removidos: ', ruidos) print('Sinal
limpo', len(sinal), 'amostras):', sinal)
```